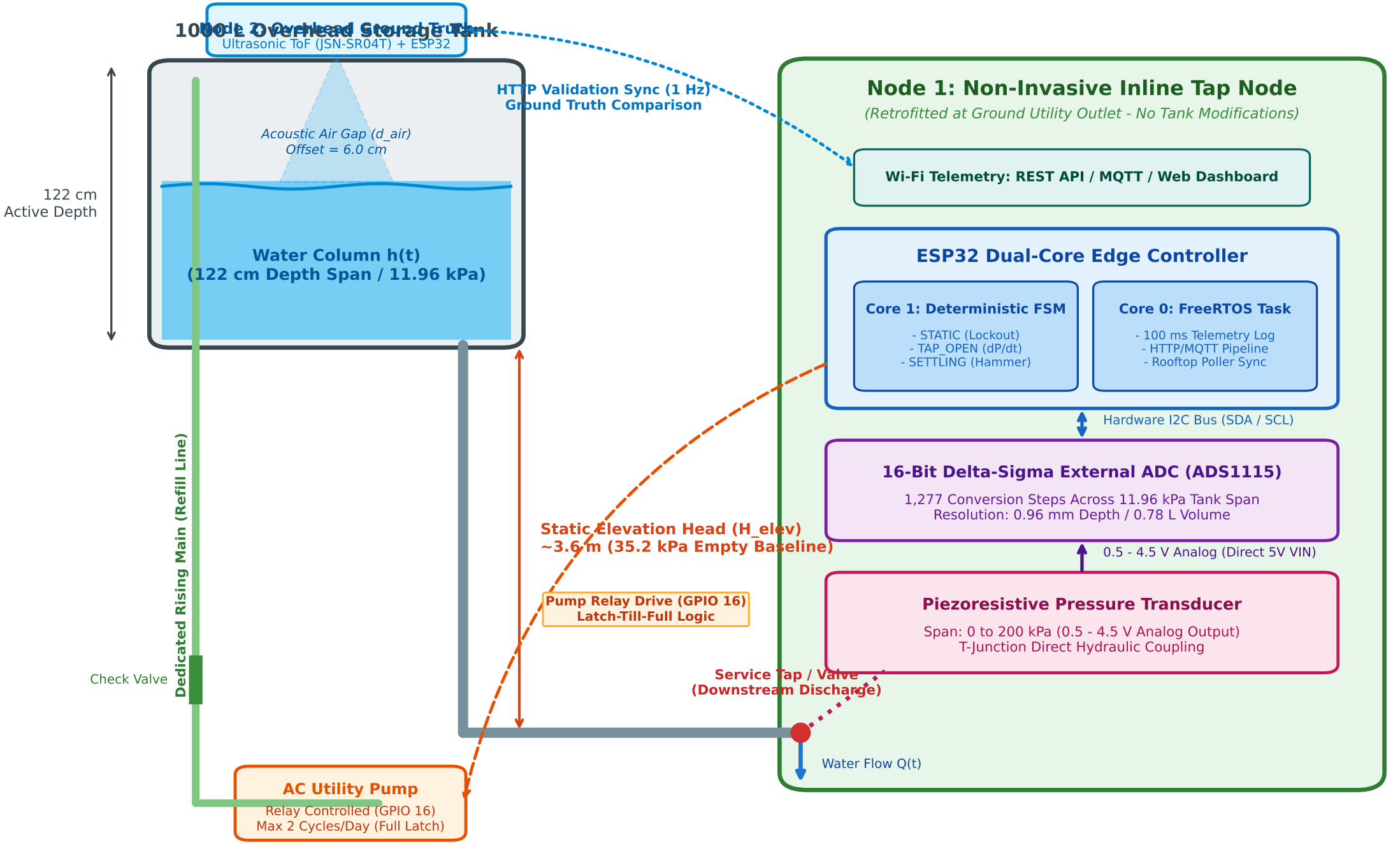

1000 L Overhead Storage Tank
Node 2: Overhead Ground Truth
Ultrasonic ToF (JSN-SR04T) + ESP32
122 cm
Active Depth
Acoustic Air Gap (d_air)
Offset = 6.0 cm
Water Column h(t)
(122 cm Depth Span / 11.96 kPa)
HTTP Validation Sync (1 Hz)
Ground Truth Comparison
Node 1: Non-Invasive Inline Tap Node
(Retrofitted at Ground Utility Outlet - No Tank Modifications)
Wi-Fi Telemetry: REST API / MQTT / Web Dashboard
ESP32 Dual-Core Edge Controller
Core 1: Deterministic FSM
- STATIC (Lockout)
- TAP_OPEN (dP/dt)
- SETTLING (Hammer)
Core 0: FreeRTOS Task
- 100 ms Telemetry Log
- HTTP/MQTT Pipeline
- Rooftop Poller Sync
Hardware I2C Bus (SDA / SCL)
16-Bit Delta-Sigma External ADC (ADS1115)
1,277 Conversion Steps Across 11.96 kPa Tank Span
Resolution: 0.96 mm Depth / 0.78 L Volume
0.5 - 4.5 V Analog (Direct 5V VIN)
Piezoresistive Pressure Transducer
Span: 0 to 200 kPa (0.5 - 4.5 V Analog Output)
T-Junction Direct Hydraulic Coupling
Static Elevation Head (H_elev)
~3.6 m (35.2 kPa Empty Baseline)
Pump Relay Drive (GPIO 16)
Latch-Till-Full Logic
Dedicated Rising Main (Refill Line)
Check Valve
Service Tap / Valve
(Downstream Discharge)
Water Flow Q(t)
AC Utility Pump
Relay Controlled (GPIO 16)
Max 2 Cycles/Day (Full Latch)